

Beam Design Technical Summary

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I. Objective of Summary

The objective of this summary is to document the design, fabrication, and experimental testing of an open cross-section T-beam configured as an indeterminate propped cantilever. The beam was designed, constructed from acrylic, and tested to validate theoretical predictions of load-bearing capacity and stiffness. Specifically, the report compares predicted and observed failure load P_{FAIL} , deflection at failure v_{FAIL} , and the effective spring constant k . The project emphasizes the relationship between beam geometry, material properties, and true structural performance.

II. Project Summary

Design of Beam

The beam was designed under the constraints of a 700 mm span (measured from clamp face to roller - 800 mm total length), a maximum cross-sectional height of 50 mm, and fabrication from a single 3 mm thick acrylic sheet. An open cross-section T-beam geometry was selected, consisting of a two-ply flange and a four-ply web to maximize bending stiffness while maintaining material efficiency. Vertical triangular stiffeners were incorporated along the beam length to improve structural rigidity and resist local deformation.

The design process relied on classical beam theory, including bending stress relationships and deflection equations, with the goal of maximizing predicted load capacity in a competitive context.

Section Properties

Section properties were calculated using the method of composite areas. The cross-section was divided into rectangular primitives, and the centroid and moment of inertia were determined using the parallel axis theorem.

Key properties:

- $I_z = 202450.7 \text{ mm}^4$
- $c_{top} = 18.74 \text{ mm}$
- $c_{bot} = 31.3 \text{ mm}$

These values were used to predict stress distribution and failure load.

Material Testing

Flexural testing of acrylic was conducted to determine material properties. The experimentally measured values were:

- $E = 2527.6 \text{ MPa}$
- $TS_f = 53.2 \text{ MPa}$

Published values are approximately:

- $E \approx 2739 \text{ MPa}$
- $TS \approx 56 \text{ MPa}$

The experimental results were within 5–8% of published values, indicating good agreement and reliable material characterization.

Fabrication and Assembly

The beam components were designed in CAD and laser cut using a BossLaser system. Assembly was completed using acrylic solvent welding, followed by clamping and curing for 24–48 hours.

During fabrication, an initial miscut of stiffener components required re-cutting using scrap material.

Testing Procedure

Two phases of testing were conducted:

Phase 1: Simply Supported Test

The beam was tested in a simply supported configuration and loaded up to approximately 500 N. This phase was used to determine the baseline stiffness.

Phase 2: Propped Cantilever Test

The beam was then reconfigured into a propped cantilever by clamping one end while maintaining a roller support at the other. A load was applied at midspan until catastrophic failure occurred.

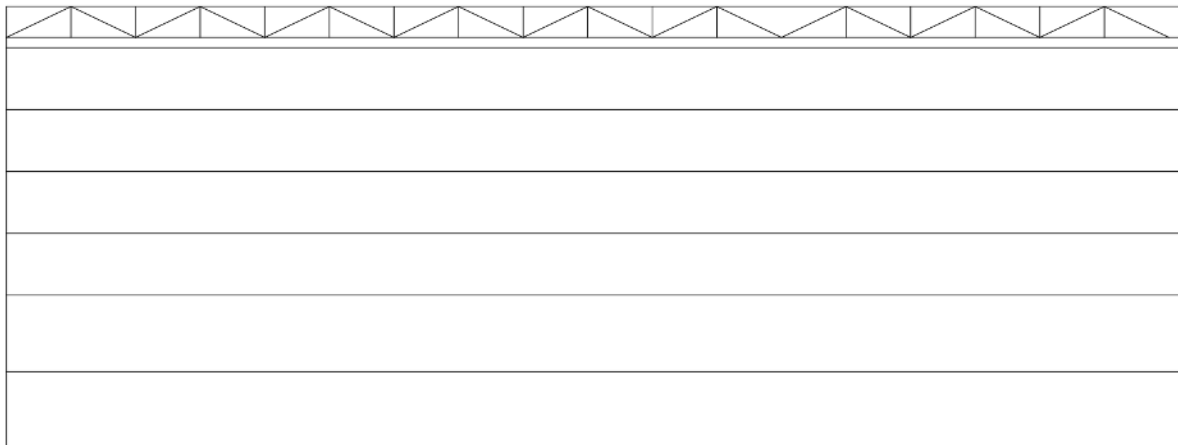


Fig. 1. Beam DXF output designed in SolidWorks.

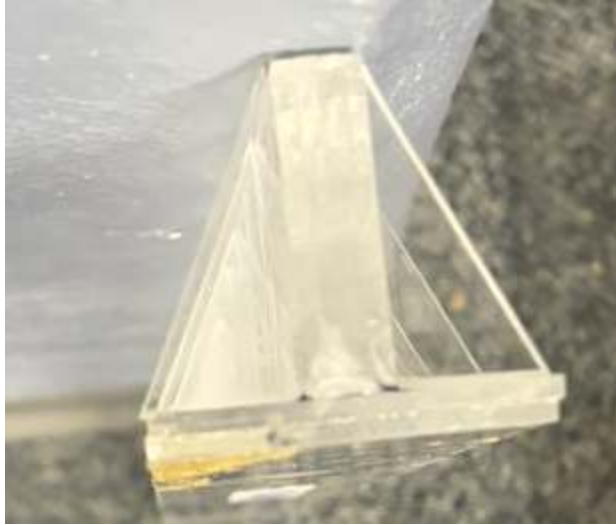


Fig. 2. Cross section of cut and solvent welded T-beam with triangle stiffeners attached at intervals.



Fig. 3. Fully constructed beam prior to testing.

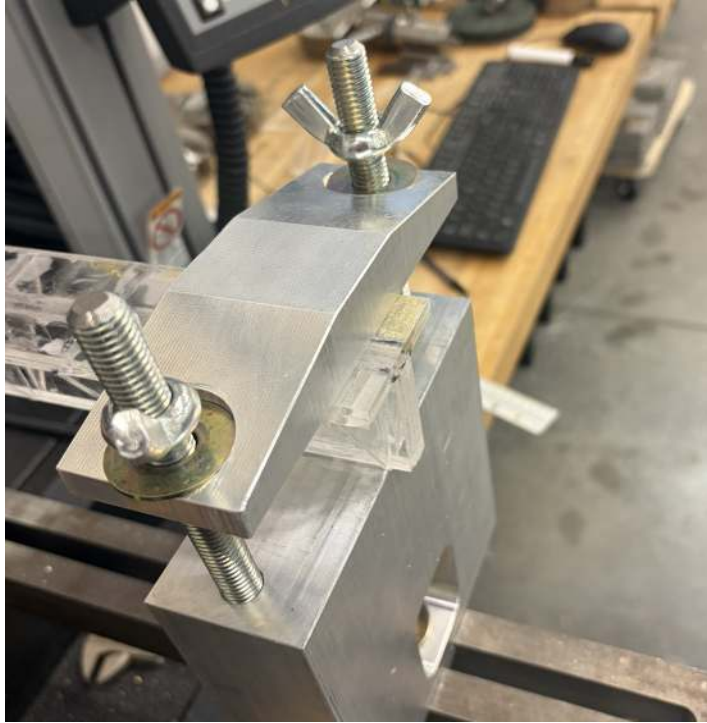


Fig. 4. Beam under loading in the simply supported configuration, used to determine baseline stiffness from the linear load–deflection response

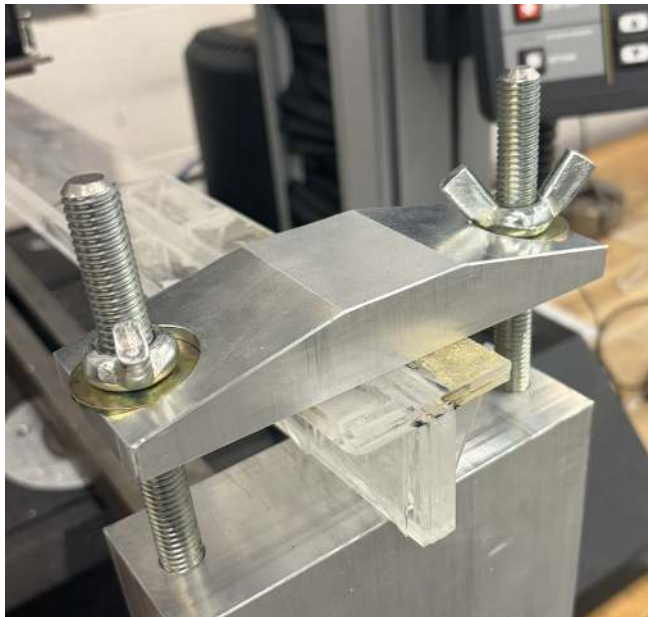


Fig. 5. Beam under loading in the propped cantilever configuration with one end clamped, demonstrating increased stiffness due to added rotational restraint.



Fig. 6. Beam after catastrophic failure under propped cantilever loading, showing brittle flexural fracture and separation into two pieces.

Experimental Results

From testing:

- $E \approx 2739 \text{ MPa}$
- $TS \approx 56 \text{ MPa}$

The beam exhibited brittle failure in flexure, with a sudden drop in load following peak force.

Table 1 - Stiffness Comparison

Configuration	k (N/mm)
Simply Supported	70.8
Propper Cantilever	111.7

The stiffness increased by approximately 58% when transitioning to the propped cantilever configuration, consistent with the added rotational constraint of the clamped support.

Table 2 - Comparison of Results

Parameter	Theoretical	Experimental	% Difference
P_{FAIL}	5419.93 N	1662.55 N	69.3 %
v_{FAIL}	23.6 mm	16.06 mm	31.9
k	229.8 N/mm	111.7 N/mm	51.4

The experimental results were consistently lower than theoretical predictions, with failure load, deflection at failure, and stiffness differing by approximately 69.3%, 31.9%, and 51.4%, respectively.